Lesson 7 Multiple PWM Servos Control

This lesson focuses on controlling the No.1 and No.2 PWM interfaces on the expansion board to achieve the motion control of multiple servos. For detailed principle instruction of PWM servo, please refer to "4.Expanded Lesson\3.Raspberry Pi Expansion Board Lesson\2.Expansion Board Control Lesson\Lesson 5 Single PWM Servo Control".

1. Working Principle

Let's view the implementation idea of this lesson:

Control the servo rotation by transmitting pulse signals and setting the rotation angle and duration of the servo through code parameters. The source code of the program is located in "/home/pi/board_demo/pwm_servo_control_demo.py".

```
if __name__ == '__main__':

    while True:
        board.pwm_servo_set_position(1, [[1, 1100]]) # 设置1号舵机脉宽为1100，运行时间为1000毫秒(set the pulse width of servo 1 to 1100 and the runtime to 1000 milliseconds)
        time.sleep(1)
        board.pwm_servo_set_position(1, [[1, 1500]]) # 设置1号舵机脉宽为1500，运行时间为1000毫秒(set the pulse width of servo 1 to 1500 and the runtime to 1000 milliseconds)
        time.sleep(1)
        board.pwm_servo_set_position(1, [[1, 1900], [2, 1000]]) #设置1号舵机脉宽为1000，设置2号舵机脉宽为1000, 运行时间为1000毫秒(set the pulse width of servo 1 and servo 2 to 1000 and the runtime to 1000 milliseconds)
        time.sleep(1)
        board.pwm_servo_set_position(1, [[1, 1500], [2, 2000]]) #设置1号舵机脉宽为2000，设置2号舵机脉宽为2000, 运行时间为1000毫秒(set the pulse width of servo 1 and servo 2 to 2000 and the runtime to 1000 milliseconds)
        time.sleep(1)
        if not start:
            board.pwm_servo_set_position(1, [[1, 1500], [2, 1500]]) # 设置1号舵机脉宽为1500，设置2号舵机脉宽为1500, 运行时间为1000毫秒(set the pulse width of servo 1 and servo 2 to 1500 and the runtime to 1000 milliseconds)
            time.sleep(1)
            print('已关闭(closed)')
            break
```

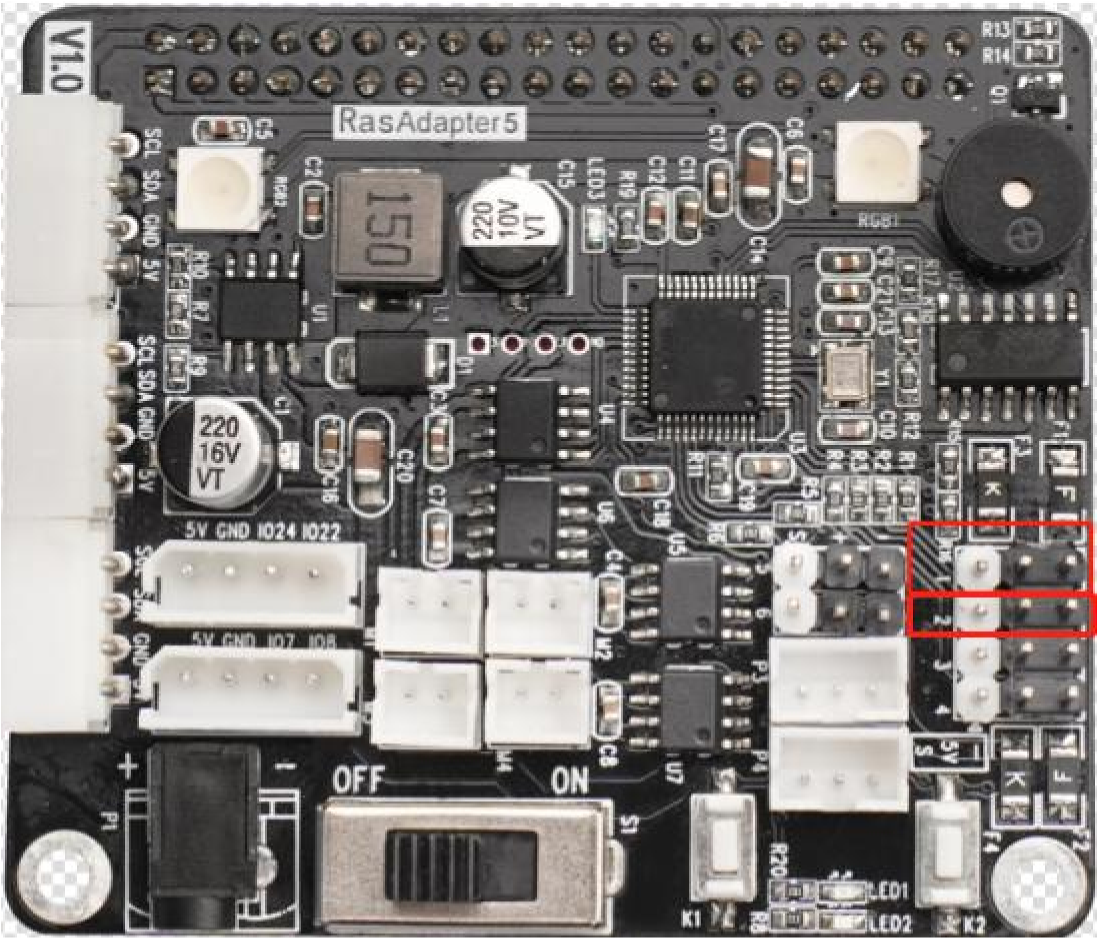
The program calls the "pwm_servo_set_position()" function in the "Board" library to implement the servo control. Take "board.pwm_servo_set_position(1, [[1, 1100]])" as an example, among them:

The first parameter "1" indicates the servo's runtime is 1s.

The second parameter "[[1, 1100]]" represents setting the No.1 servo's pulse width as 1100.

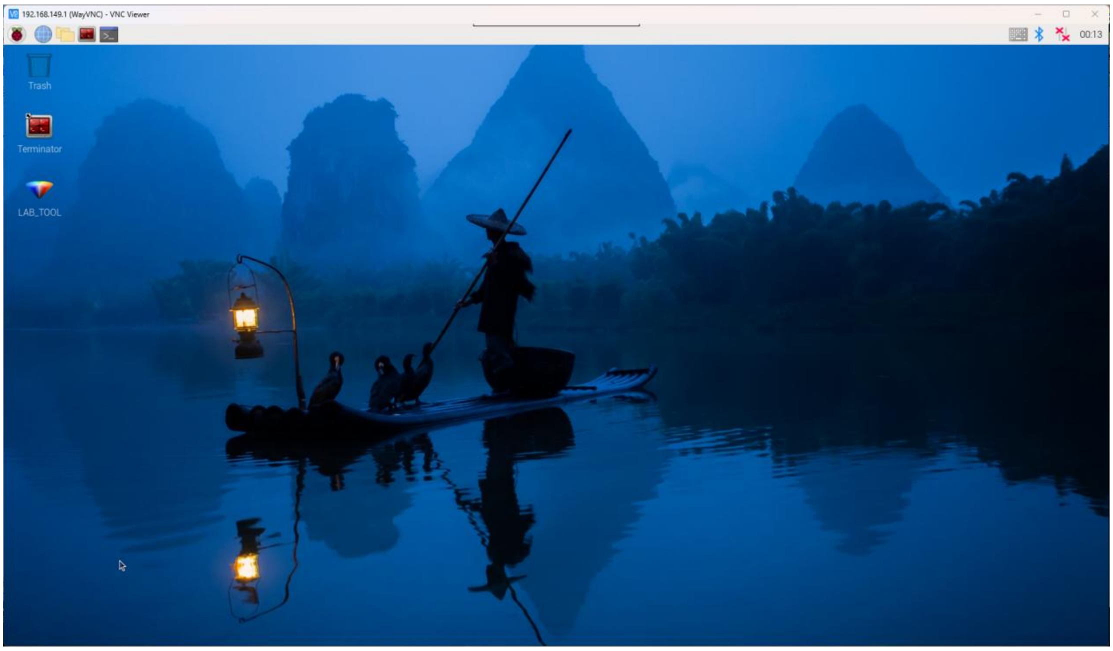
2. Getting Ready

There are 6 PWM servo interfaces on the Raspberry Pi expansion board. Let's test by connecting the No.1 and No.2 servo interfaces, as shown below:



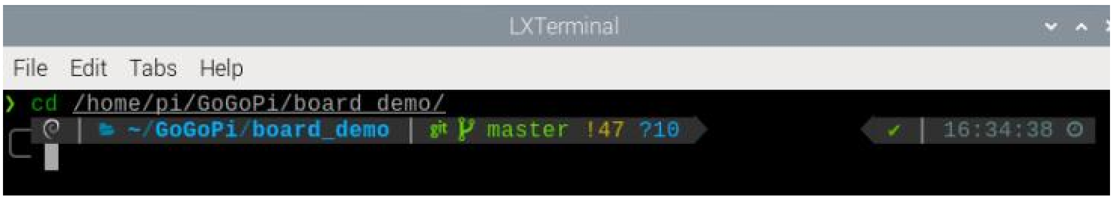
3. Operation Steps

1) Turn the device on and connect the robot through the VNC remote connection tool.



2) Click  in the top left corner of the desktop, or press the shortcut key “Ctrl+Alt+T” to open the command line terminal.

3) Enter the command “cd board_demo/”. Then press “Enter” to navigate to the directory where the demo program is located.



4) Enter the command “python3 pwm_servo_control_demo.py”. Press “Enter” to execute the program.

5) To close the program, press “Ctrl+c”. If the program cannot be closed successfully, repeat this operation until it exits.

4. Program Outcome

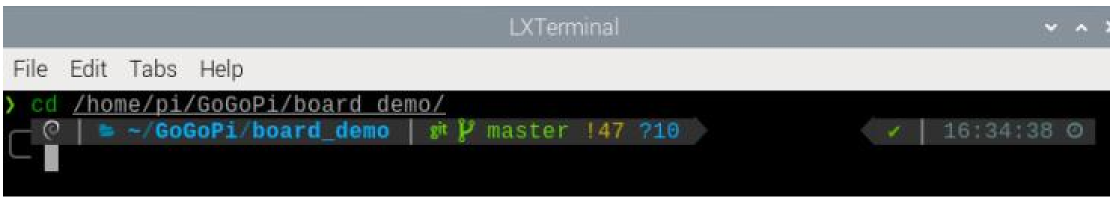
Once the program is executed, the servos 1 and 2 rotate circularly.

5. Function Extension

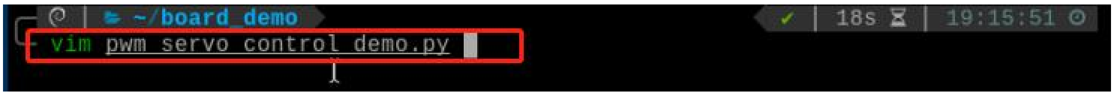
5.1 Replace Servo Interface

The program sets to rotate the servo connected to No.1 servo interface first by default. If it’s necessary to replace the servo interface, insert the servo into the specified interface first, then perform the operation. Here is an example of changing to No.2 servo interface:

1) Enter the command “cd /home/pi/board_demo/”. Then press “Enter” to navigate to the directory where the demo program is located.

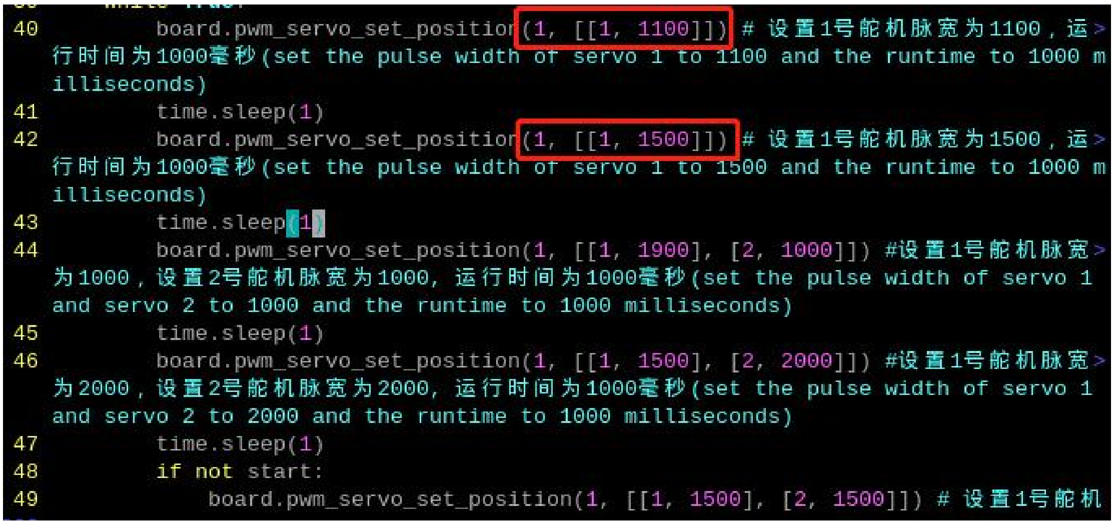


2) Enter the command “vim pwm_servo_control_demo.py”. Press “Enter” to open the program file.

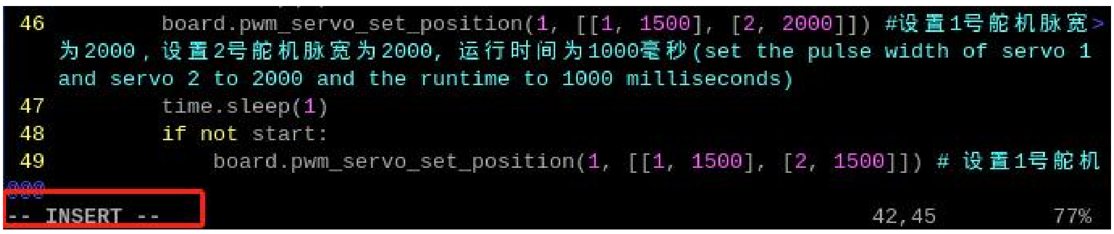


3) Enter “:set number” in the opened page to display the line numbers. Please set it based on your own needs.

4) Locate the codes shown below in the opened page:



5) Press “i” on the keyboard to enter the editing mode.



6) Modify the first parameters “1” into “2” in parentheses of the “board.pwm_servo_set_position”, as shown below:

```
    while True:
        board.pwm_servo_set_position(1, [[2, 1100]]) # 设置1号舵机脉宽为1100，运行时间为1000毫秒(set the pulse width of servo 1 to 1100 and the runtime to 1000 milliseconds)
        time.sleep(1)
        board.pwm_servo_set_position(1, [[2, 1500]]) # 设置1号舵机脉宽为1500，运行时间为1000毫秒(set the pulse width of servo 1 to 1500 and the runtime to 1000 milliseconds)
        time.sleep(1)
        board.pwm_servo_set_position(1, [[1, 1900], [2, 1000]]) #设置1号舵机脉宽为1000，设置2号舵机脉宽为1000, 运行时间为1000毫秒(set the pulse width of servo 1 and servo 2 to 1000 and the runtime to 1000 milliseconds)
        time.sleep(1)
        board.pwm_servo_set_position(1, [[1, 1500], [2, 2000]]) #设置1号舵机脉宽为2000，设置2号舵机脉宽为2000, 运行时间为1000毫秒(set the pulse width of servo 1 and servo 2 to 2000 and the runtime to 1000 milliseconds)
        time.sleep(1)
```

7) After the modification is completed, press “Esc” on the keyboard. Then enter “:wq” (do not miss the colon before wq), and press “Enter” to save and exit the program.

```
        board.pwm_servo_set_position(1, [[1, 1500], [2, 2000]]) #设置1号舵机脉宽为2000，设置2号舵机脉宽为2000, 运行时间为1000毫秒(set the pulse width of servo 1 and servo 2 to 2000 and the runtime to 1000 milliseconds)
        time.sleep(1)
        if not start:
            board.pwm_servo_set_position(1, [[1, 1500], [2, 1500]]) # 设置1号舵机
:wq
```

8) After it exits, enter the command “python3 pwm_servo_control_demo. Py” to see the modified effect.

5.2 Modify Rotation Angle

The program sets the initial state of servo 1 to rotate in a loop between 54 and 90 degrees. If you want to customize the servo rotation angle, for example, change the servo to rotate in a loop from 30 to 120 degrees, please follow the steps below.

1) Follow the steps in “5.1 Replace Servo Interface” to open the program file, and locate the codes shown below.

```
if __name__ == '__main__':

    while True:
        board.pwm_servo_set_position(1, [[1, 1100]]) # 设置1号舵机脉宽为2500，运行时间为1000毫秒
        time.sleep(1)
        board.pwm_servo_set_position(1, [[1, 1500]]) # 设置1号舵机脉宽为1500，运行时间为1000毫秒
```

2) The servo rotation range is between 500 and 2500 pulse widths, which is equivalent to 0 to 180 degrees. For example, 1500 pulse width is 90 degrees, which means that 1 degree is equal to 11.1 pulse widths. The formula for converting angles to position values is 11.1 * angle + 500.

```
    while True:
        board.pwm_servo_set_position(1, [[1, 833]]) # 设置1号舵机脉宽为1100，运行时间为1000毫秒(set the pulse width of servo 1 to 1100 and the runtime to 1000 milliseconds)
        time.sleep(1)
        board.pwm_servo_set_position(1, [[1, 1832]]) # 设置1号舵机脉宽为1500，运行时间为1000毫秒(set the pulse width of servo 1 to 1500 and the runtime to 1000 milliseconds)
        time.sleep(1)
        board.pwm_servo_set_position(1, [[1, 1900], [2, 1000]]) #设置1号舵机脉宽为1000，设置2号舵机脉宽为1000, 运行时间为1000毫秒(set the pulse width of servo 1 and servo 2 to 1000 and the runtime to 1000 milliseconds)
        time.sleep(1)
```

3) After the modification is completed, press “Esc” on the keyboard. Then enter “:wq” (do not miss the colon before wq), and press “Enter” to save and exit the program.

```
        board.pwm_servo_set_position(1, [[1, 1500], [2, 2000]]) #设置1号舵机脉宽为2000，设置2号舵机脉宽为2000, 运行时间为1000毫秒(set the pulse width of servo 1 and servo 2 to 2000 and the runtime to 1000 milliseconds)
        time.sleep(1)
        if not start:
            board.pwm_servo_set_position(1, [[1, 1500], [2, 1500]]) # 设置1号舵机
@@@
:wq
```

4) After it exits, enter the command “python3 pwm_servo_control_demo. Py” to see the modified effect.